# Supplementary File S4: Direct-Node Process Mission Summaries

This file gives the parameter and secondary-measurement details for the four deterministic direct-node missions summarized in the manuscript. Each row represents one simulator trajectory with a fixed seed. The missions used different process models and were not repeated trials or matched controls.

Table 1: Recorded actions, endpoint observations, and evidence limits.

| Scene | Recorded sequence and endpoint | Interpretation limit |
|---|---|---|
| Injection molding | Target 32.5 ± 0.35 g. Holding pressure changed from 45 to 57.73 bar and holding time from 6 to 10.56 s; a later pressure request reached 62 bar. Final part weight was 32.36 g. | One trajectory with multiple parameter changes; effects cannot be separated. |
| Film extrusion | Target thickness 25.0 ± 0.7 $\mu$m. Four requests changed casting speed, fast-roll speed, rail width, and casting speed again. Thickness changed from 28.03 to 25.30 $\mu$m. Recorded melt temperatures stayed within 268–300°C. | No repeated trajectory or matched controller; no causal attribution to an individual request. |
| Annealing | Target hardness 95 ± 6 HV. A zone-2 temperature request changed 710 to 850°C, and speed changed from 140 to 152 m/min. Hardness observations were 99.18 and 99.74 HV. A 164 m/min candidate was reverted after event text reported 101.0 HV; the mission summary reported 99.5 HV at the final 152 m/min setting. | No measured throughput gain. Final tensile-strength and surface-defect readings were not archived. |
| Wastewater | Target dissolved oxygen (DO) was 2.8–4.0 mg/L. Five blower increases changed frequency from 26 to 34.1 Hz; the structured DO sequence rose from 0.194 to 3.567 mg/L. Total phosphorus changed from 0.662 to 0.5507, 0.4607, and 0.413 mg/L after subsequent dosing. The mission summary reported final DO of 3.54 mg/L and a simulator cost index of 180.2, down from 182.6. | The index is simulator-defined, not measured operating cost. The complete final effluent vector was not archived. |

These summaries do not supply complete raw trajectories or independent replications. The endpoints support the reported mission outcomes only; they do not establish the causal effect of each control action, a repeatable process benefit, or performance on physical equipment.